



Earth System

Earth System Science



Chapter 4: The Climate Change



Objective

- *To Explain Climate as a system*
- *To Know Climate Change and its Dynamics*
- *To Identify The Cause of Climate Change*

CHAPTER 4

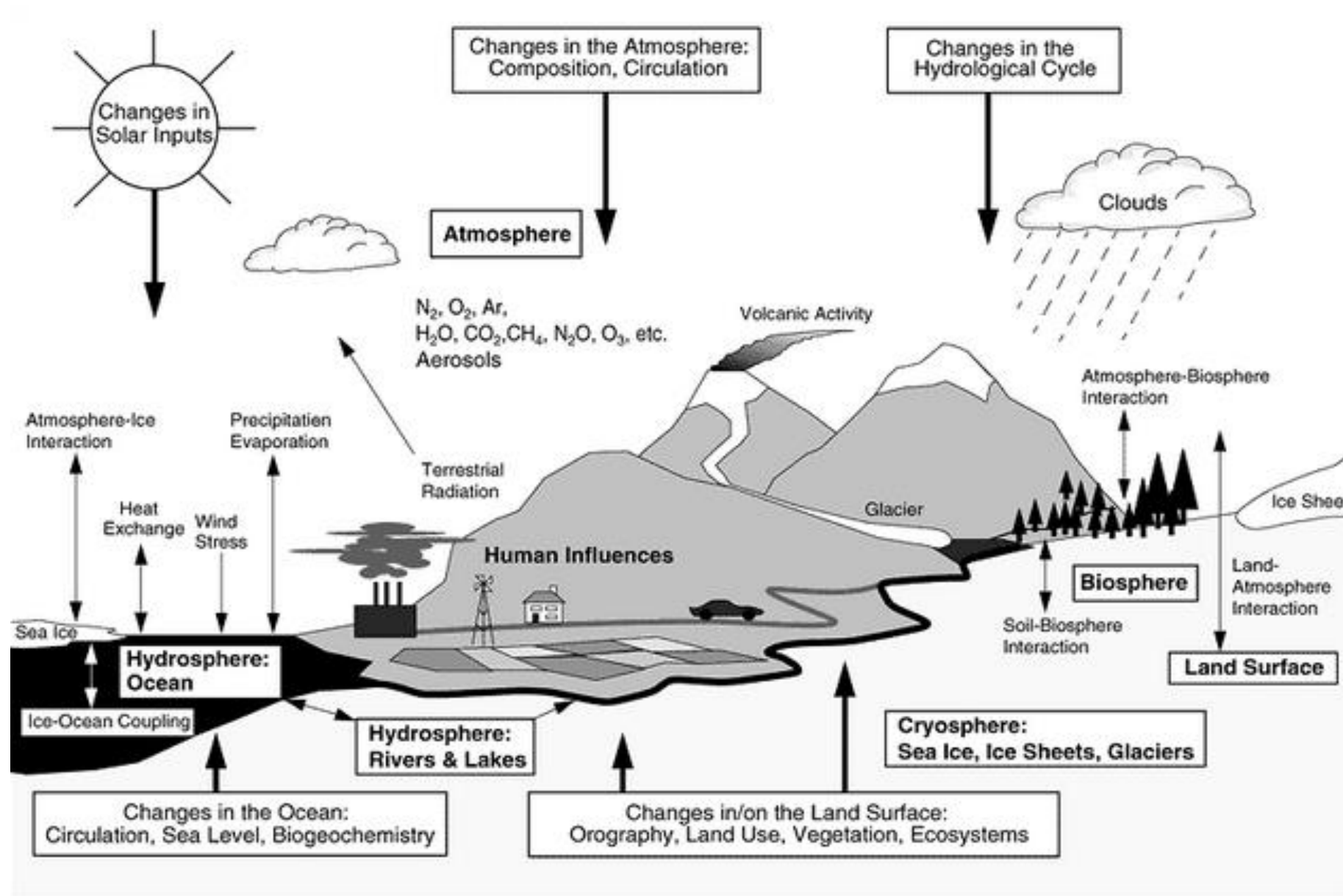
THE CLIMATE CHANGE

The Climate System

Its components

- The climate system is an interactive system consisting of all the four earth system spheres: the atmosphere, the hydrosphere, the biosphere and the lithosphere
- It is forced or influenced by various external forcing mechanisms, the most important of which is **the Sun**.
- Also the **direct effect of human activities** on the climate system is considered as an external forcing.

- Human activities are generally considered **external forcing** in the context of the climate system because they introduce new factors that influence climate patterns from outside the natural system's equilibrium.
- While they originate from within the Earth system (i.e., humans are part of the biosphere), their impacts on the climate—such as greenhouse gas emissions, deforestation, and aerosol production—are treated as **external forcings** because they alter the natural energy balance of the atmosphere and climate dynamics.



Schematic view of the components of the global climate system (bold), their processes and interactions (thin arrows) and some aspects that may change (bold arrows).

Atmosphere in the climate system

- It is the most unstable and rapidly changing part of the system
- It contains greenhouse gases: carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O)
- the atmosphere also contains aerosols and clouds, which interact with the incoming and outgoing radiation in a complex and spatially very variable manner.

- Aerosols are a suspension of airborne solid or liquid particles, with a typical size between a few nanometers and 10 μm , that reside in the atmosphere for at least several hours.
- Many aerosols act as surfaces for water droplets and ice crystals to form on.
- The most variable component of the atmosphere is water in its various phases such as vapor, cloud droplets, and ice crystals.

Hydrosphere in the climate system

- Runoff from the land returning to the ocean influences the ocean's composition and circulation.
- Oceans store and transport a large amount of energy
- Oceans dissolve and store great quantities of carbon dioxide (CO₂).
- Because of their high heat capacity, they damp vast and strong temperature changes and function as a regulator of the Earth's climate
- And as a source of natural climate variability, in particular on the longer time-scales: e.g. ENSO

- Glaciers and snow fields operate in the climate system with their high reflectivity (albedo) for solar radiation
- Because the ice sheets store a large amount of water, variations in their volume are a potential source of sea level variations

Biosphere in the climate system

- The marine and terrestrial biospheres have a major impact on the atmosphere's composition: via carbon cycle, nitrogen cycle
- Through the photosynthetic process, both marine and terrestrial plants (especially forests) store significant amounts of carbon from carbon dioxide.
- Microbes in natural wetlands produce methane gas.
- Combustion of fossil fuels release CO₂
- Nitrous oxide (N₂O) gas is produced when nitrate and ammonium in fertilizers breakdown in the soil.

Lithosphere in the climate system

- Soil is the second largest carbon store, or ‘sink’, after the oceans.
- Soil is a living environment for the bacteria and microbes that are involved in biogeochemical cycles
- Surface roughness influences the atmosphere dynamically as winds blow over the land’s surface.
Roughness is determined by both topography and vegetation.
- Wind also blows dust from the surface into the atmosphere: *dust is an aerosol.*

Climate change and its Dynamics

First, WHAT IS CLIMATE?

- The average weather,
- or more strictly, the statistical description in terms of the ***mean and variability of relevant quantities*** over a period of time ranging from months to thousands or millions of years.
- The classical period for averaging these variables is 30 years, as defined by the World Meteorological Organization.
- The relevant quantities are most often surface variables such as *temperature, precipitation and wind*.

- Weather can change quickly, from one moment to the next and over short distances.

*It can be raining one minute, and snowing the next.
It can be pouring on one side of town and sunny on the other.*

- Climate, on the other hand, changes more slowly.
- Also, climate generally doesn't vary much over short distances, except in the mountains.

WHAT IS CLIMATE CHANGE?

- *Intergovernmental Panel on Climate Change (IPCC)*
definition: “Climate change’ refers to a change in the state of the climate that can be identified (e.g., using statistical tests) by **changes in the mean** and/or **the variability** of its properties, and that **persists for an extended period**, typically decades or longer”
- It refers to any change in climate over time, whether due to natural variability or as a result of human activity.

- This usage differs from that in the United Nations Framework Convention on Climate Change (UNFCCC), where climate change refers to a change of climate that is attributed **directly or indirectly to human activity** that alters the composition of the global atmosphere
- and that is in addition to natural climate variability observed over comparable time periods.

The Cause of Climate Change

- Climate change may be due to internal processes and/or external forcing:

Internal processes

refers to natural fluctuations within the climate system that occur due to interactions between components like the atmosphere, oceans, land, and ice. **Examples:**

- *El Niño/La Niña*
- *Ocean Circulation*
- *Volcanic activity*

External forcing

are factors outside the climate system that drive long-term climate changes. These can be **natural** or **human-induced**.

❖ **Natural external forcings**

- Changes in solar radiation
- Orbital changes (Milankovitch cycles)

❖ **Human-induced external forcings**

- Greenhouse gas emissions
- Land use changes:

- There are two fundamentally important ingredients of the dynamics of the climate.

These are:

Radiative forcing: the causing agent

Feedbacks in the climate system: the play makers

Radiative forcing

What is Radiative FORCING?

- Radiative forcing is the net change in the energy balance of the Earth system due to some imposed perturbation.
- It is the difference between insolation (sunlight) absorbed by the Earth and energy radiated back to space
- Positive radiative forcing means Earth receives more incoming energy from sunlight than it radiates to space, which produces warming
- Conversely, negative radiative forcing means that Earth loses more energy to space than it receives from the sun, which produces cooling.

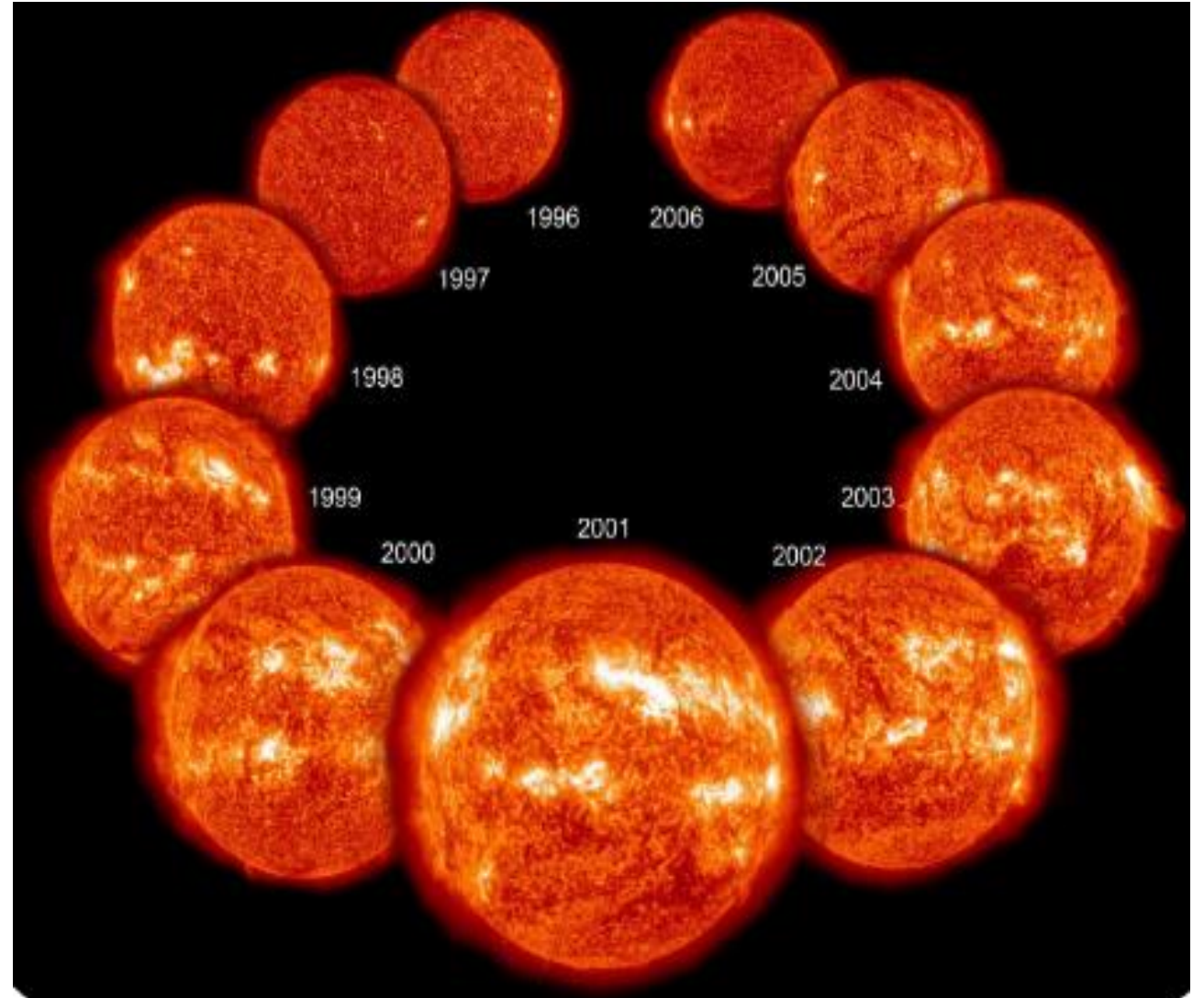
- Greenhouse gases, such as carbon dioxide (CO₂), methane (CH₄), and water vapor, contribute to positive radiative forcing by trapping heat in the atmosphere.
- Aerosols can have both positive (warming) and negative (cooling) radiative forcing effects, depending on their composition and interactions with clouds.

- Radiative forcing represents any external factor that influences global climate by heating or cooling the planet.
- They may be either natural or anthropogenic.
- **Natural** radiative forcings include ***volcanic eruptions, solar variations*** and ***orbital forcing***.
- **Anthropogenic** radiative forcing include ***changes in the composition of the atmosphere*** the release of greenhouse gases and aerosols , ***land use change***.

Natural Radiative Forcings

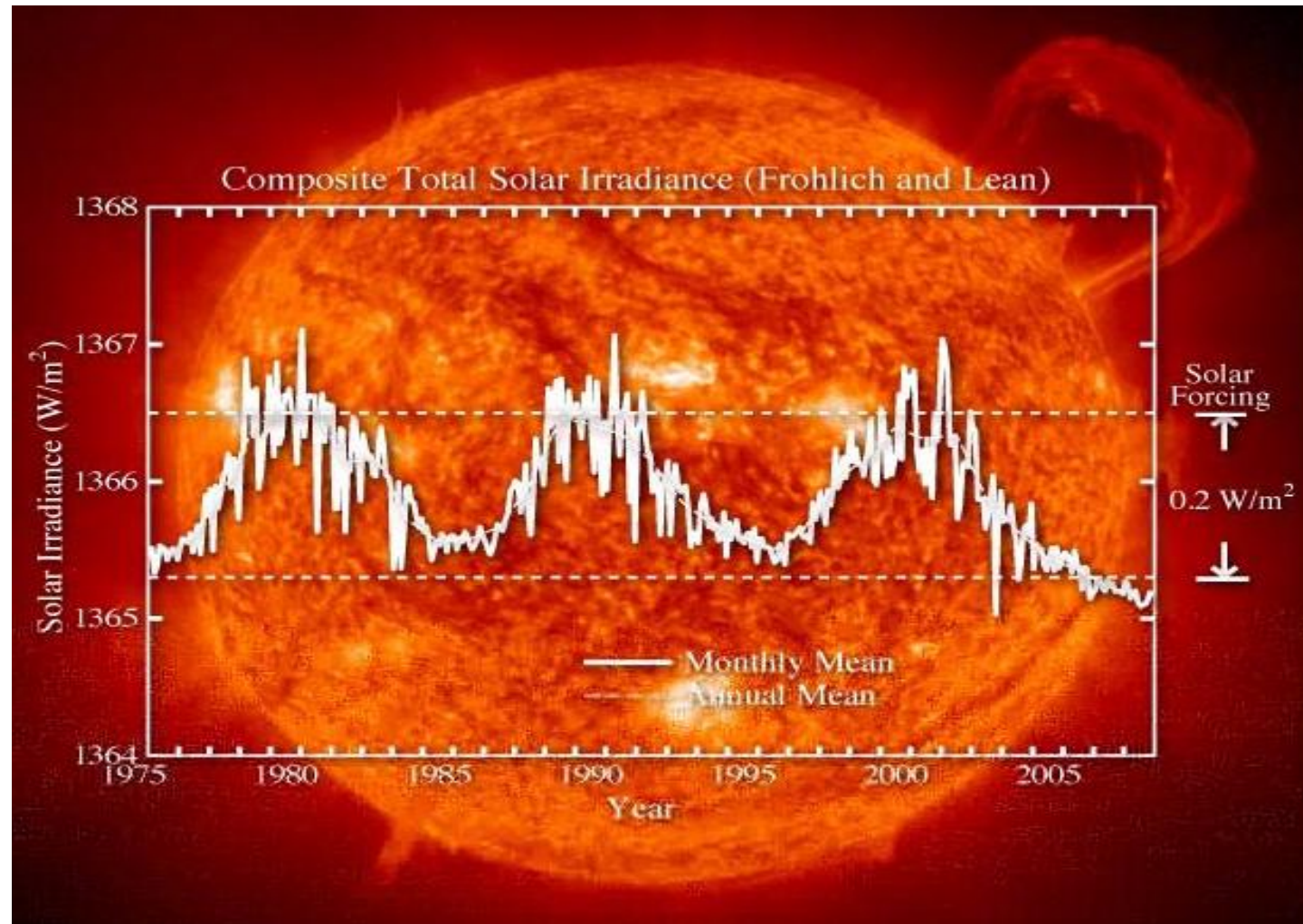
- **Solar variations** are changes in the amount of radiant energy emitted by our Sun due to solar cycle
- The **solar cycle** is the nearly periodic 11-year change in the Sun's activity: *including changes in the levels of solar radiation and ejection of solar material and appearance*
- Every 11 years or so, the Sun's north and south poles switch places. Then it takes about another 11 years for the Sun's north and south poles to flip back again.

- The beginning of a solar cycle is a **solar minimum**, or when the Sun has the least sunspots.
- The middle of the solar cycle is the **solar maximum**, or when the Sun has the most sunspots.
- As the cycle ends, it fades back to the solar minimum and then a new cycle begins.



Images of the Sun during one solar cycle.

Total solar irradiance over the past three solar cycles, since 1975, varying between 1365 and 1367 W/m².

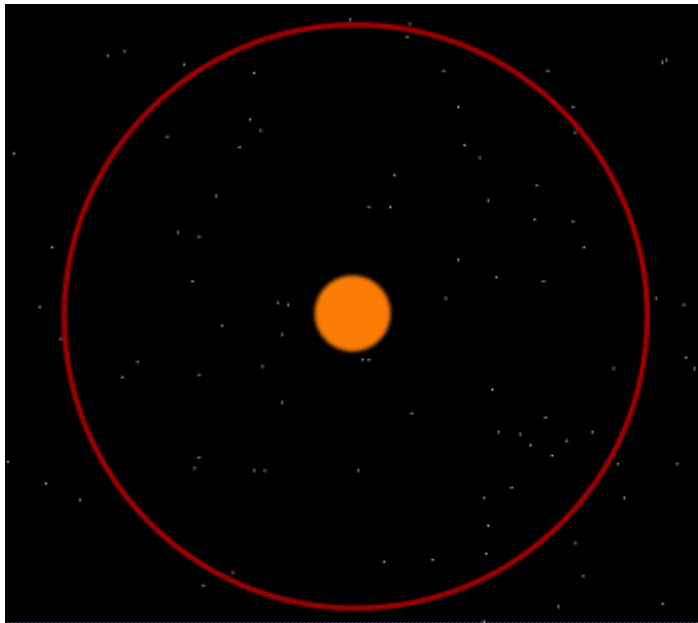


The solar irradiance is the output of light energy from the entire disk of the Sun, measured at the Earth.

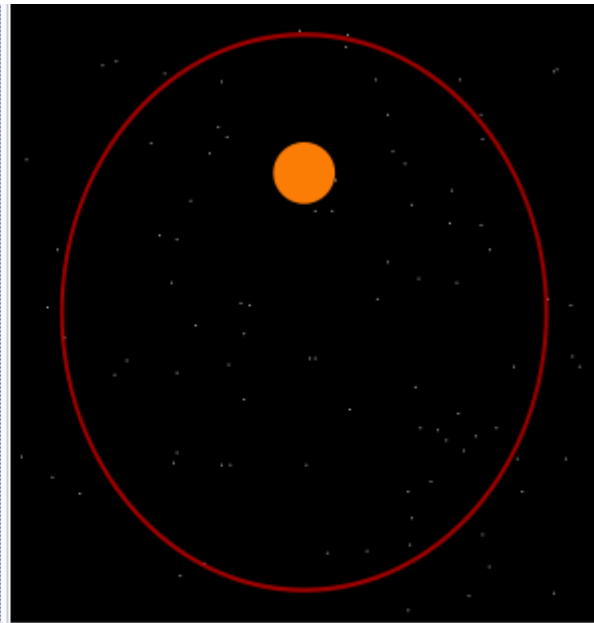
Cycle	Years	Max Year	Cycle	Years	Max Year
1	1755–1766	1761	14	1902–1913	1906
2	1766–1775	1769	15	1913–1923	1917
3	1775–1784	1778	16	1923–1933	1928
4	1784–1798	1788	17	1933–1944	1937
5	1798–1810	1805	18	1944–1954	1947
6	1810–1823	1816	19	1954–1964	1958
7	1823–1833	1829	20	1964–1976	1968
8	1833–1843	1837	21	1976–1986	1979
9	1843–1856	1848	22	1986–1996	1989
10	1856–1867	1860	23	1996–2008	2001
11	1867–1878	1870	24	2008–2019	2014
12	1878–1890	1883	25	2019–~2030	2024–25
13	1890–1902	1894			

- **Orbital forcing** refers to the effect of slow changes in the tilt of the Earth's axis and shape of the orbit on the climate
- Orbital changes change the total amount of sunlight reaching the Earth
- A theory connecting orbital changes to climatic changes is the **Milankovitch theory**
- Milankovitch cycle describes variations in ***eccentricity, axial tilt, and precession*** of the Earth's orbit influence climatic patterns on Earth through orbital forcing.

- The **orbital eccentricity (e) (Shape of Earth's Orbit)** of the earth determines the amount by which its orbit around the sun deviates from a perfect circle.



Circular orbit, no eccentricity



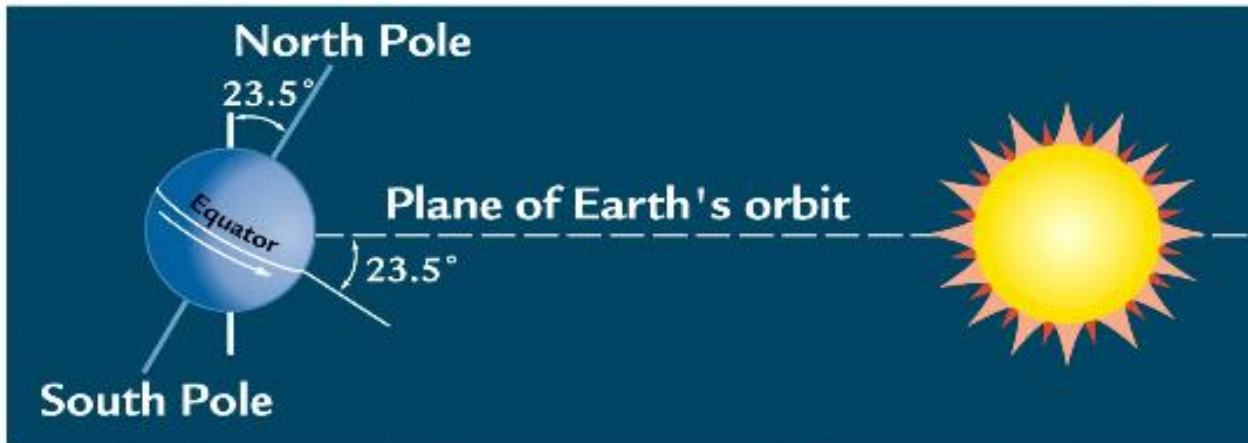
Orbit with 0.5 eccentricity; Earth's orbit is never this eccentric

For the earth's orbit to stretch from nearly a circular to an elliptical orbit and back again takes nearly 100,000 years.

circular orbit: $e = 0$ elliptic orbit: $0 < e < 1$

- **Axial tilt (obliquity)** is the angle between earth's rotational axis and its orbital axis.
- Earth's obliquity oscillates between 22.1 and 24.5 degrees on a 41,000-year cycle.

Tilt (Obliquity)

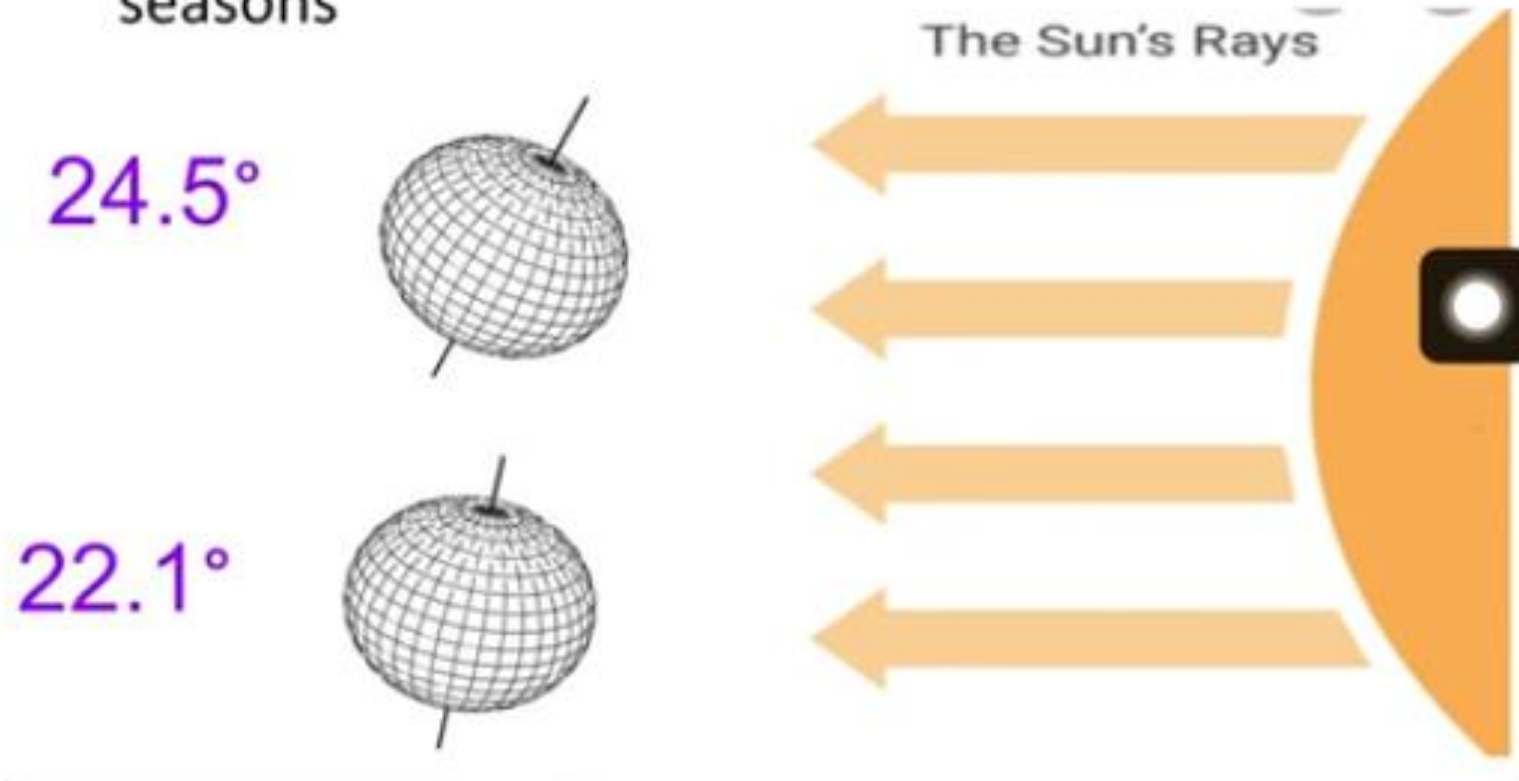


Effect:

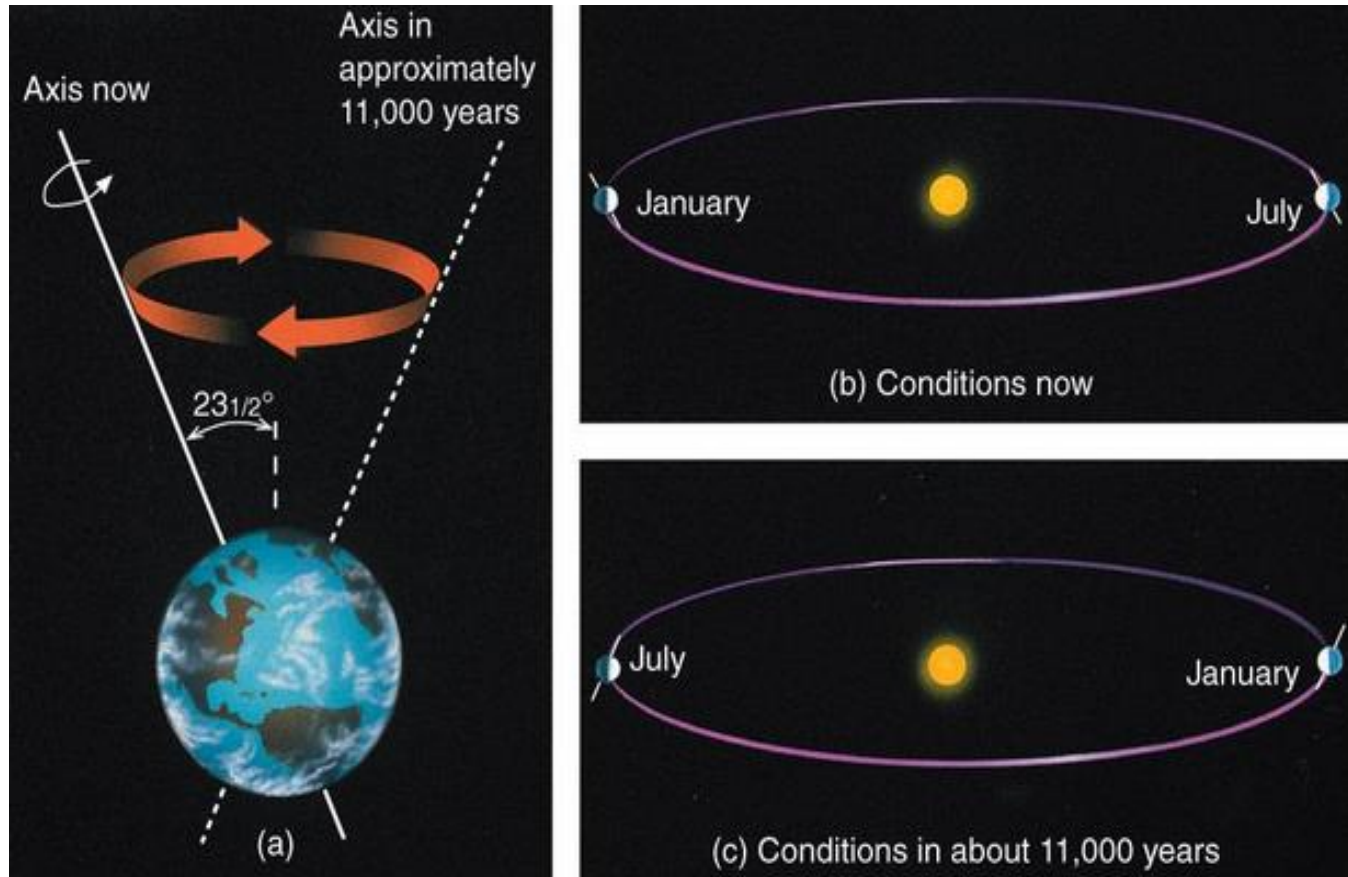
- Greater tilt → more extreme seasons (hotter summers, colder winters).
- Smaller tilt → milder seasons

#1: Tilt of Earth's Axis

- When the Earth is MORE tilted towards the sun, the Earth will have more EXTREME seasons



- **Precession** is the change in orientation of the Earth's rotational axis



- The change in the orientation is tracked based on fixed stars
- One full cycle of precession takes about 26 years

- Precession changes the timing of when Earth reaches perihelion (closest to the Sun) and aphelion (farthest from the Sun). This shifts the seasons in relation to Earth's elliptical orbit.

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Thanks !

Earth System Science



Chapter 4: The Climate Change



Objective

- To Explain Anthropogenic radiative forcing
- To Know Feedbacks in the climate system

Anthropogenic Radiative Forcing

Anthropogenic radiative forcing

- Forcing due to human, rather than natural, factors.
- Such factors include increased greenhouse gas concentrations associated with **fossil fuel burning**,
Humans have substantially increased atmospheric greenhouse gases

“it is extremely likely that human activities caused more than half of the observed increase in global mean surface temperatures from 1951 to 2010.” IPCC

Key Components of Anthropogenic Radiative Forcing

Greenhouse Gases (GHGs):

- ✓ Human activities, such as burning fossil fuels, deforestation, and industrial processes, release GHGs like carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O).
- ✓ These gases trap outgoing longwave radiation in the atmosphere, leading to a positive radiative forcing and warming of the Earth's surface.
- ✓ Example: CO₂ concentrations have increased significantly since the Industrial Revolution, enhancing the greenhouse effect.

Aerosols:

- ❑ Human activities release aerosols (tiny particles) from sources like burning fossil fuels and biomass.
- ❑ Aerosols can have both cooling and warming effects:
 - Cooling effect: Aerosols scatter and reflect sunlight back to space, leading to negative radiative forcing (e.g., sulfate aerosols).
 - Warming effect: Certain aerosols, like black carbon (soot), absorb sunlight and heat the atmosphere.

Land-Use Changes:

- Human-induced changes in land surface properties among other things.

Human activity has modified the land cover and changed the surface albedo.

- ☐ Activities like deforestation, urbanization, and agriculture alter the Earth's surface properties (e.g., albedo, or reflectivity).
- ☐ Deforestation reduces carbon sequestration
- ☐ Human activity has modified the land cover and changed the surface albedo
- ☐ Urban areas with dark surfaces absorb more energy compared to vegetated surfaces.

Ozone (O₃):

- ❑ Human emissions of pollutants like nitrogen oxides (NO_x) and volatile organic compounds (VOCs) have increased **tropospheric ozone**, which acts as a greenhouse gas and contributes to warming.
- ❑ However, depletion of **stratospheric ozone** due to chlorofluorocarbons (CFCs) causes cooling in the stratosphere.

- Land use change, and in particular *deforestation*, also has significant impacts on greenhouse gas concentration
- Land use change can also impact the climate through modifications in the *surface roughness*, *latent heat flux* and *river runoff*.
- Also humans generate direct input of heat to the atmosphere by consuming energy by consuming energy for *industrial, residential, and transportation purposes*

- Human activity may change the water cycle through

- Irrigation via

- ❖ Evapotranspiration Increases

- ❖ Altered Runoff and Infiltration:

- ❖ Groundwater Depletion

- ❖ Regional Humidity Changes

- power plant cooling

Thermal power plants (e.g., coal, nuclear, natural gas) use large amounts of water for cooling, often withdrawing water from nearby rivers, lakes, or seas.

- ✓ Increased Evaporationse

- ✓ Thermal Pollution

- ✓ Water Withdrawals

❑ Anthropogenic radiative forcing is a major driver of global warming and climate change. by altering the energy balance of the Earth system, it affects:

- Temperature increases
- Changes in precipitation patterns
- Sea-level rise
- Ecosystem shifts

❑ Mitigating anthropogenic radiative forcing involves reducing greenhouse gas emissions, limiting aerosol pollution, and adopting sustainable land-use practices.

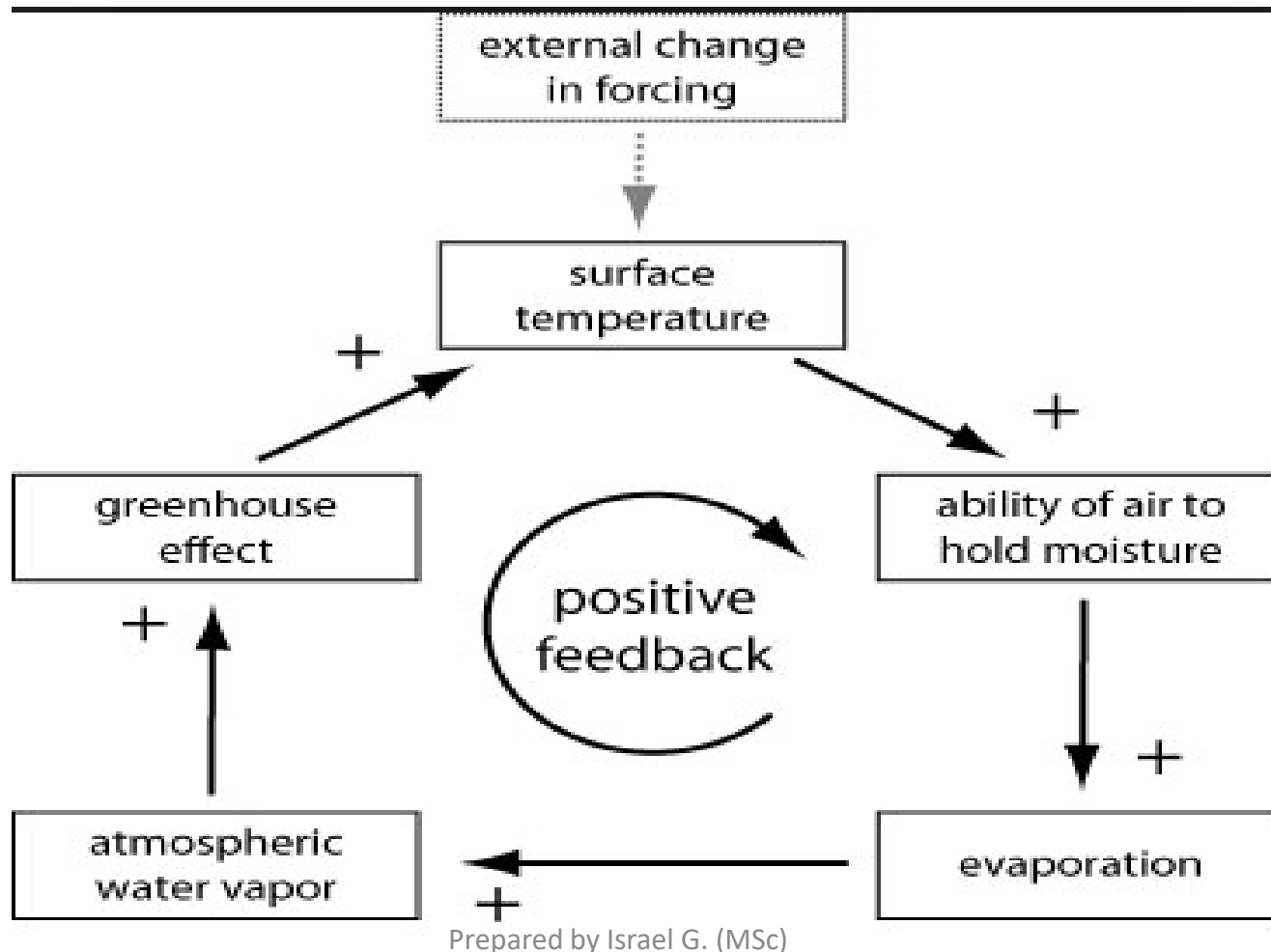
Feedbacks in the climate system

Feedbacks in the climate system

- Feedback in climate is An interaction in which a perturbation (change) in one climate quantity causes a change in a second, and the change in the second quantity ultimately leads to an additional change in the first.
- *A negative feedback* is one in which the initial perturbation is weakened by the changes it causes
- Whereas, *a positive feedback* is one in which the initial perturbation is increased.

Key feedbacks in the climate system

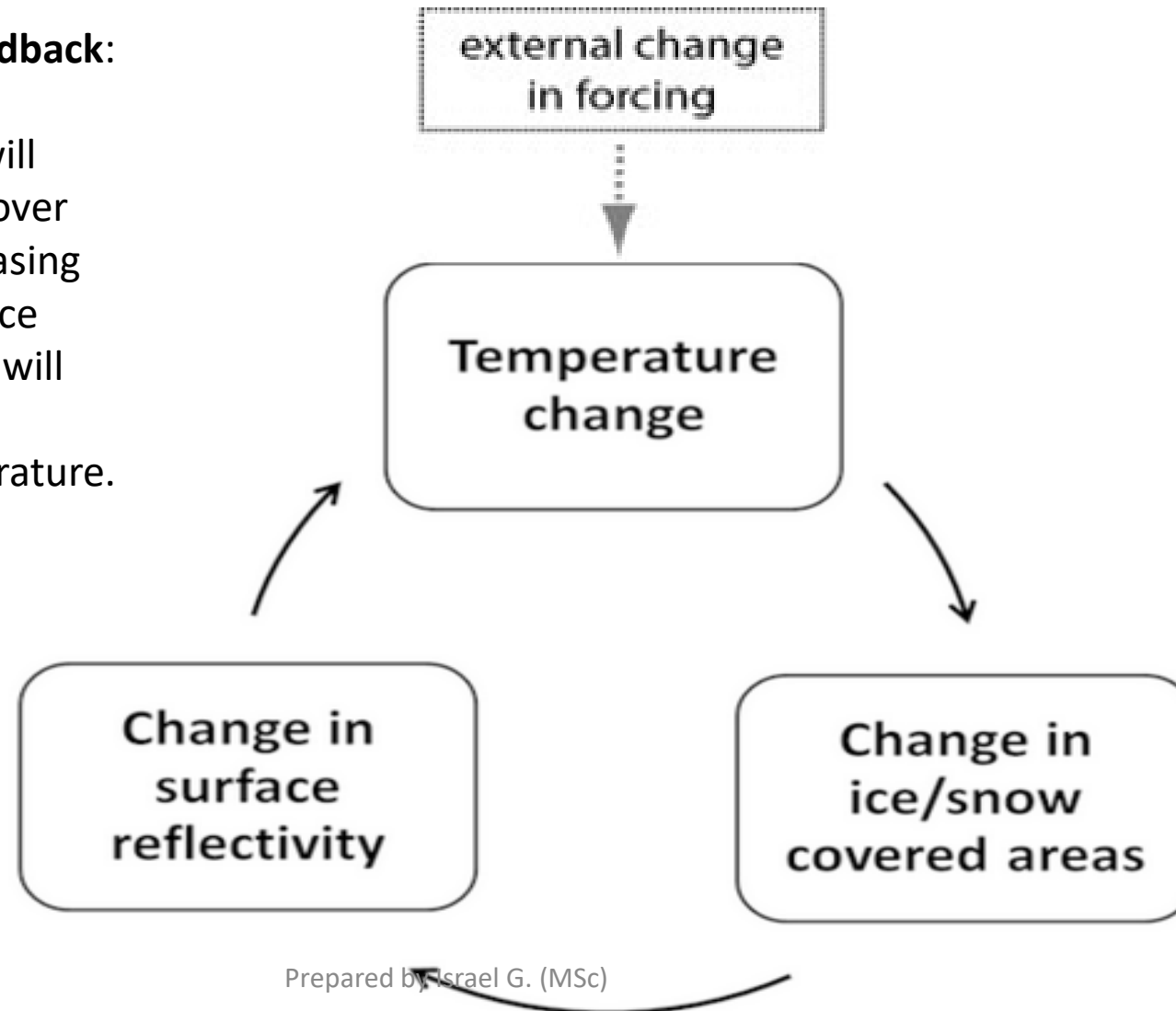
I. Water vapor feedback



II. Ice-albedo feedback

In positive feedback:

Increase in temperature will decrease ice cover thereby decreasing reflected surface radiation. This will intern amplify surface temperature.



- ❑ Positive feedback loops can potentially *lead to more significant* and rapid changes in the climate system, while negative feedbacks can *act as stabilizing mechanisms*.
- ❑ The complex interplay between radiative forcing and feedbacks contributes to the dynamic nature of the Earth's climate.
- ❑ Scientists study these interactions to *improve climate models* and *enhance our understanding* of how the climate system responds to various influences.

- Understanding The Interaction Between Radiative Forcing And Feedbacks Is Crucial For
 - Predicting The Magnitude And Nature Of Climate Change.
 - Quantifying Climate Sensitivity
 - Improving Climate Models
 - Assessing Policy Impacts

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Thanks !